P2 Chapter 9: Differentiation

Differentiating Sin and Cos

Differentiating trigonometric functions

$$\frac{d}{dx}(\sin x) = \mathbf{\cos x}$$

Things of helpfulness:

- As $x \to 0$, $\sin x \approx x$ and $\cos x \approx 1 - \frac{1}{2}x^2$
- $\sin(a+b) = \sin a \cos b + \cos a \sin b$

Proof

Yes, you could be tested on this!

If $y = f(x)$ then $\frac{dy}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$

$$= \lim_{h \to 0} \frac{\sin(x+h) - \sin(x)}{h}$$

$$= \lim_{h \to 0} \frac{\sin(x)\cos(h) + \cos(x)\sin(h) - \sin(x)}{h}$$

Use addition formula.

$$= \lim_{h \to 0} \left(\left(\frac{\cos h - 1}{h} \right) \sin x + \left(\frac{\sin h}{h} \right) \cos x \right)$$

Collect $\sin x$ terms and $\cos x$ terms together.

Since $\lim_{h \to 0} \sin h \approx h$, $\therefore \lim_{h \to 0} \frac{\sin h}{h} \approx \lim_{h \to 0} \frac{h}{h} = 1$

Similarly, $\lim_{h \to 0} \frac{\cos h - 1}{h} \approx \lim_{h \to 0} \frac{1 - \frac{1}{2}h^2 - 1}{h} = \lim_{h \to 0} (-\frac{1}{2}h) \to 0$

Use small angle approximations to consider what the expressions in terms of h will become.

$$\therefore \frac{dy}{dx} = \lim_{h \to 0} (0 \sin x + 1 \cos x) = \cos x$$

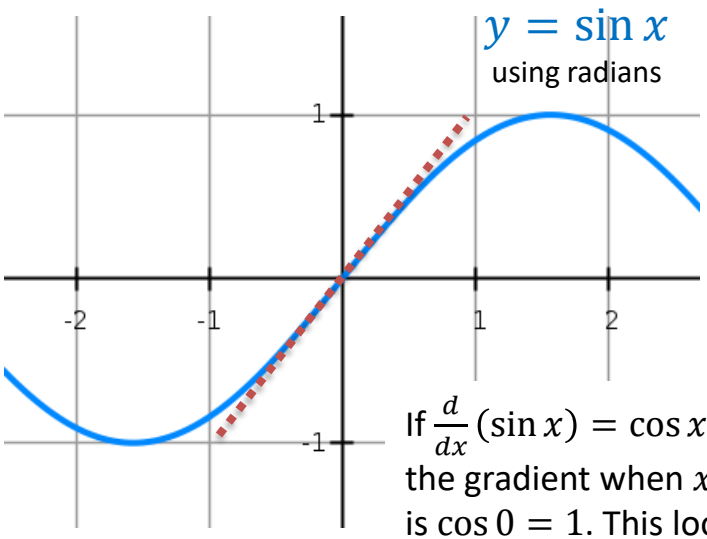
Why does this result only hold in radians?

Explanation 1:

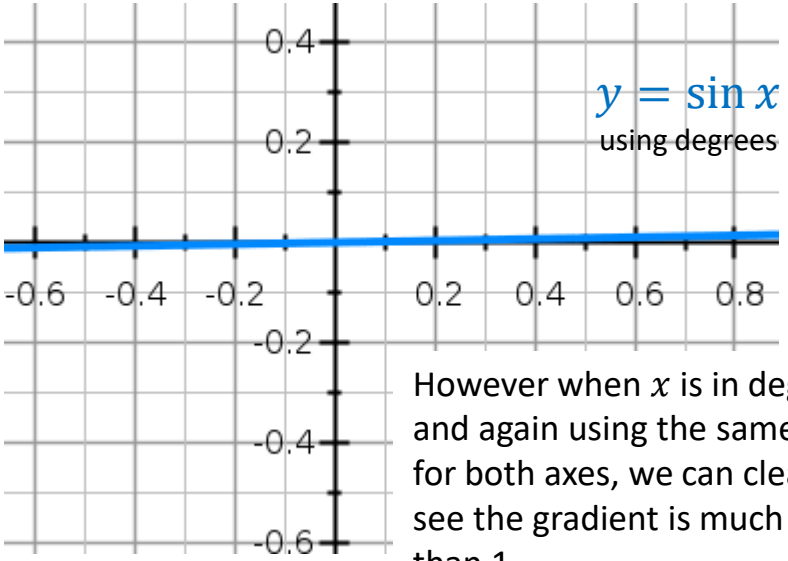
The approximations $\sin x \approx x$ and $\cos x \approx 1 - \frac{1}{2}x^2$ (as $x \to 0$) only holds if x is in radians (we saw why in the chapter on radians). The proof that $\frac{d}{dx}(\sin x) = \cos x$ made use of these approximations.

Explanation 2:

We can see by observation if we look at the graph of $\sin x$ in radians and in degrees.



If $\frac{d}{dx}(\sin x) = \cos x$ then the gradient when $x = 0$ is $\cos 0 = 1$. This looks right from the graph.



However when x is in degrees, and again using the same scale for both axes, we can clearly see the gradient is much lower than 1.

Differentiating trigonometric functions

$$\frac{d}{dx}(\cos x) = -\mathbf{\sin x}$$

Memory Tip: The way I picture this in my head is this:

$$\begin{array}{c} \sin x \\ \downarrow \\ \cos x \end{array}$$

where moving 'down' is differentiating and moving up is integrating. If it's the 'wrong' function (differentiating cos or integrating sin) the sign changes.

Differentiating $\sin kx$ and $\cos kx$

$$\frac{d}{dx}(\sin kx) = \boldsymbol{k \cos kx}$$

$$\frac{d}{dx}(\cos kx) = \boldsymbol{-k \sin kx}$$

Note: This is not a rule in itself but a specific case of the '*chain rule*' which we'll see later this chapter.

Quickfire Questions:

$\frac{d}{dx}(\sin 3x) =$?

$\frac{d}{dx}(\cos 5x) =$?

$\frac{d}{dx}(3 \sin 5x) =$?

$\frac{d}{dx}(4 \cos 3x) =$?

$\frac{d}{dx}\left(-\frac{1}{2}\sin x\right) =$?

$\frac{d}{dx}\left(-\frac{2}{3}\cos \frac{1}{2}x\right) =$?

Differentiating $\sin kx$ and $\cos kx$

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Note: This is not a rule in itself but a specific case of the '*chain rule*' which we'll see later this chapter.

Quickfire Questions:

$$\frac{d}{dx}(\sin 3x) = \boldsymbol{3 \cos 3x}$$

$$\frac{d}{dx}(\cos 5x) = \boldsymbol{-5 \sin 5x}$$

$$\frac{d}{dx}(3 \sin 5x) = \boldsymbol{15 \cos 5x}$$

$$\frac{d}{dx}(4 \cos 3x) = \boldsymbol{-12 \sin 3x}$$

$$\frac{d}{dx}\left(-\frac{1}{2}\sin x\right) = \boldsymbol{-\frac{1}{2}\cos x}$$

$$\frac{d}{dx}\left(-\frac{2}{3}\cos\frac{1}{2}x\right) = \boldsymbol{\frac{1}{3}\sin\frac{1}{2}x}$$

Example

[Textbook] A curve has equation $y = \frac{1}{2}x - \cos 2x$. Find the stationary points on the curve in the interval $0 \leq x \leq \pi$.

?

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$$\frac{dy}{dx} = \frac{1}{2} + 2\sin 2x = 0$$

$$2\sin 2x = -\frac{1}{2}$$

$$\sin 2x = -\frac{1}{4}$$

$$2x = -0.2527, 3.3943, 6.0305$$

$$x = 1.697, \qquad 3.015$$

When $x = 1.697,\ y = \frac{1}{2}(1.697) - \cos(2 \times 1.697) = 1.82$

When $x = 3.015,\ y = 0.539$

Test Your Understanding

A curve has equation $y = \sin 3x + 2x$. Find the stationary points on the curve in the interval $0 \leq x \leq \frac{2}{3}\pi$.

?

Test Your Understanding

A curve has equation $y = \sin 3x + 2x$. Find the stationary points on the curve in the interval $0 \leq x \leq \frac{2}{3}\pi$.

$$\frac{dy}{dx} = 3cos\ 3x + 2 = 0$$
$$3\cos 3x = -2$$
$$\cos 3x = -\frac{2}{3}$$
$$3x = 2.3005, 3.9827$$
$$x = 0.7668, \qquad 1.3276$$

When $x = 0.7668,\ y = 2.279$

When $x = 1.3276,\ y = 1.910$

Exercise 9.1

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Homework Exercise

1 **a** Given that $f(x) = \cos x$, show that

$$f'(x) = \lim_{h \to 0}\left(\left(\frac{\cos h - 1}{h}\right)\cos x - \frac{\sin h}{h}\sin x\right)$$

b Hence prove that $f'(x) = -\sin x$.

Problem-solving

Use the definition of the derivative and the addition formula for $\cos(A + B)$.

2 Differentiate:

a $y = 2\cos x$ **b** $y = 2\sin\frac{1}{2}x$ **c** $y = \sin 8x$ **d** $y = 6\sin\frac{2}{3}x$

3 Find $f'(x)$ given that:

a $f(x) = 2\cos x$ **b** $f(x) = 6\cos\frac{5}{6}x$ **c** $f(x) = 4\cos\frac{1}{2}x$ **d** $f(x) = 3\cos 2x$

4 Find $\frac{dy}{dx}$ given that:

a $y = \sin 2x + \cos 3x$ **b** $y = 2\cos 4x - 4\cos x + 2\cos 7x$

c $y = x^2 + 4\cos 3x$ **d** $y = \frac{1 + 2x\sin 5x}{x}$

5 A curve has equation $y = x - \sin 3x$. Find the stationary points of the curve in the interval $0 \leqslant x \leqslant \pi$.

6 Find the gradient of the curve $y = 2\sin 4x - 4\cos 2x$ at the point where $x = \frac{\pi}{2}$

7 A curve has the equation $y = 2\sin 2x + \cos 2x$. Find the stationary points of the curve in the interval $0 \leqslant x \leqslant \pi$.

Homework Exercise

8 A curve has the equation $y = \sin 5x + \cos 3x$. Find the equation of the tangent to the curve at the point $(\pi, -1)$. **(4 marks)**

9 A curve has the equation $y = 2x^2 - \sin x$. Show that the equation of the normal to the curve at the point with x-coordinate π is

$$x + (4\pi + 1)y - \pi(8\pi^2 + 2\pi + 1) = 0$$

(7 marks)

10 Prove, from first principles, that the derivative of $\sin x$ is $\cos x$.

You may assume the formula for $\sin(A + B)$ and that as $h \to 0$, $\frac{\sin h}{h} \to 1$ and $\frac{\cos h - 1}{h} \to 0$.

(5 marks)

Challenge

Prove, from first principles, that the derivative of $\sin(kx)$ is $k\cos(kx)$.

You may assume the formula for $\sin(A + B)$ and that as $h \to 0$, $\frac{\sin kh}{h} \to k$ and $\frac{\cos kh - 1}{h} \to 0$.

Homework Answers

1 **a** $f'(x) = \lim_{h \to 0}\left(\frac{f(x+h) - f(x)}{h}\right)$

$$= \lim_{h \to 0}\left(\frac{\cos(x+h) - \cos x}{h}\right)$$

$$= \lim_{h \to 0}\left(\frac{\cos x \cos h - \sin x \sin h - \cos x}{h}\right)$$

$$= \lim_{h \to 0}\left(\frac{\cos x(\cos h - 1) - \sin x \sin h}{h}\right)$$

$$= \lim_{h \to 0}\left(\left(\frac{\cos h - 1}{h}\right)\cos x - \left(\frac{\sin h}{h}\right)\sin x\right)$$

b As $h \to 0$, $\cos h \to 1$, so $\left(\frac{\cos h - 1}{h}\right) \to 0$

and $\left(\frac{\sin h}{h}\right) \to 1$

So $f'(x) = \lim_{h \to 0}\left(\left(\frac{\cos h - 1}{h}\right)\cos x - \left(\frac{\sin h}{h}\right)\sin x\right)$

$$= 0 \cos x - \sin x = -\sin x$$

2 **a** $-2\sin x$ **b** $\cos\left(\frac{1}{2}x\right)$

c $8\cos 8x$ **d** $4\cos\left(\frac{2}{3}x\right)$

3 **a** $-2\sin x$ **b** $-5\sin\left(\frac{5}{6}x\right)$

c $-2\sin\left(\frac{1}{2}x\right)$ **d** $-6\sin 2x$

4 **a** $2\cos 2x - 3\sin 3x$ **b** $-8\sin 4x + 4\sin x - 14\sin 7x$

c $2x - 12\sin 3x$ **d** $-\frac{1}{x^2} + 10\cos 5x$

5 (0.41, −0.532), (1.68, 2.63), (2.50, 1.56)

6 8 **7** (0.554, 2.24), (2.12, −2.24)

8 $y = -5x + 5\pi - 1$

9 $\frac{dy}{dx} = 4x - \cos x$

At $x = \pi$, $y = 2\pi^2$, $\frac{dy}{dx} = 4\pi - \cos\pi = 4\pi + 1$

Gradient of normal $= -\frac{1}{4\pi + 1}$

Equation of normal:

$$y - 2\pi^2 = -\frac{1}{4\pi + 1}(x - \pi)$$

$$(4\pi + 1)y - 2\pi^2(4\pi + 1) = -x + \pi$$

$$x + (4\pi + 1)y - 8\pi^3 - 2\pi^2 - \pi = 0$$

$$x + (4\pi + 1)y - \pi(8\pi^2 + 2\pi + 1) = 0$$

10 Let $f(x) = \sin x$

$$f'(x) = \lim_{h \to 0}\frac{f(x+h) - f(x)}{h} = \frac{\sin(x+h) - \sin x}{h}$$

$$= \lim_{h \to 0}\frac{\sin x \cos h + \cos x \sin h - \sin x}{h}$$

$$= \lim_{h \to 0}\left[\left(\frac{\cos h - 1}{h}\right)\sin x + \left(\frac{\sin h}{h}\right)\cos x\right]$$

Since $\frac{\cos h - 1}{h} \to 0$ and $\frac{\sin h}{h} \to 1$ the expression inside the limit $\to (0 \times \sin x + 1 \times \cos x)$

So $\lim_{h \to 0}\frac{\sin(x+h) - \sin x}{h} = \cos x$

Hence the derivative of $\sin x$ is $\cos x$.

Homework Answers

Challenge

Let $f(x) = \sin(kx)$

$$f'(x) = \lim_{h \to 0}\left(\frac{f(x+h) - f(x)}{h}\right) = \lim_{h \to 0}\left(\frac{\sin(kx + kh) - \sin kx}{h}\right)$$

$$= \lim_{h \to 0}\left(\frac{\sin kx \cos kh + \cos kx \sin kh - \sin kx}{h}\right)$$

$$= \lim_{h \to 0}\left(\left(\frac{\cos kh - 1}{h}\right)\sin kx + \left(\frac{\sin kh}{h}\right)\cos kx\right)$$

As $h \to 0$, $\left(\frac{\sin kh}{h}\right) \to k$ and $\left(\frac{\cos kh - 1}{h}\right) \to 0$ as given,

So $f'(x) = 0 \sin kx + k \cos kx = k \cos kx$